

Persistent homology analysis of biomolecular data

By Guo-Wei Wei

The technological advance in the past few decades has fueled the exponential growth of omic data in biology. An important mission of biology is to understand the rules of life from existing omic data sets, which offers unprecedented opportunities for mathematicians. One of major rules of life is biomolecular structure-function relationship. Understanding such a relationship is the holy grail of biophysics and a central issue in experimental biology.

Geometric modeling is vital to the understanding of biomolecular structure-function relationships, and bridges the gap between biological data and theoretical models, such as quantum mechanics, molecular mechanics, statistical mechanics, thermodynamics, and multiscale models. However, geometry based models are often inundated with too much structural detail and are frequently computationally intractable. Topology provides the ultimate abstraction of geometric complexity by concerning only the connectivity of different components in a space and characterizing independent entities, rings and higher dimensional faces of the space in terms of topological invariants or Betti numbers. To study topological invariants in a discrete data set, such as atoms in a biomolecule, algebraic topology utilizes simplicial complexes under various settings, such as the Vietoris-Rips complex, Čech complex or alpha complex. Specifically, a 0-simplex is a vertex, a 1-simplex an edge, a 2-simplex a triangle, and a 3-simplex a tetrahedron, as illustrated in Fig. 1. Algebraic groups built on these simplicial complexes are used in simplicial homology to systematically compute Betti numbers for the given data set [1].

Nevertheless, traditional topology and homology are truly free of metrics or coordinates, and thus keep too little geometric information to be practically useful for biomolecules. Persistent homology, a new branch of algebraic topology, embeds multiscale geometric information into topological invariants to achieve an interplay between geometry and topology [2]. It creates a variety of topological spaces of a given object by varying a filtration parameter, such as the radius of balls or the level set of a real-valued function. As a result, persistent homology can capture topological features continuously over a range of spatial scales. Persistent homology analysis is often visualized by barcodes [3] or persistence diagrams [4]. As such, the changes of topological invariants over scales are recorded by the “birth”, “death” and “persistence” of barcodes over filtration. Persistent homology has been applied to a variety of domains, including image/signal analysis, chaotic dynamics, sensor networks, complex networks, shape recognition and computational biology [5].

For nano- and biomolecules, persistent homology enables a quantitative topological analysis via topological fingerprints (TFs) [6, 7]. Such an analysis reveals biomolecular “topology-function relationships”. Contrary to the commonly held belief, short-lived topological events are not noise, but are part of TFs, playing a valuable role in the quantitative topological analysis of protein folding stability [6] and fullerene curvature energy [8]. Differential geometry was utilized to derive partial differential equation (PDE) based persistence for biomolecules [8]. Multidimensional persistence induced by a multiresolution analysis [9] is particularly useful for resolving ill-posed inverse problems in cryo-EM structure determination [10].

TFs provide biomolecules a systematic and entirely different representation that cannot be literally cast into traditional physical interpretation. Fortunately, this representation is ideally suited for machine learning, particularly deep learning, which captures nonlinear and high-order interactions among features in sufficiently large and intrinsically complex data sets. One of the first integrations of machine learning and TFs offered encouraging classification of tens of thousands of proteins involving hundreds of tasks [11]. However, it turns out that persistent homology neglects chemical and biological information during the topological simplification and thus is not as competitive as geometry or physics based representation in quantitative predictions. Element specific persistent homology or multicomponent persistent homology built on colored biomolecular networks has

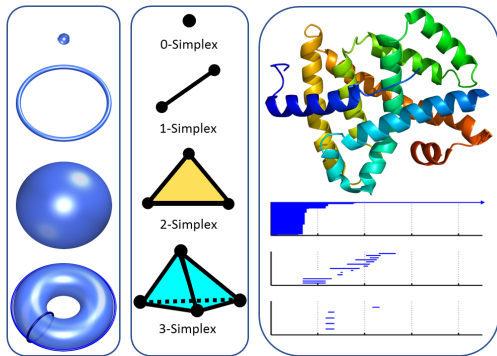


Figure 1: An illustration of topological invariants (Left column), basic simplexes (Middle column) and protein persistent barcodes (Right column). Left column: A point, a circle, an empty sphere and a torus are displayed from top to bottom. Betti-0, Betti-1 and Betti-2 numbers are, respectively, 1, 0 and 0 for a point, 0, 1 and 0 for a circle, 0, 0 and 1 for a sphere, and 1, 2 and 1 for a torus. Two auxiliary rings are added to the torus to explain Betti-1= 2. Middle column: Four typical simplexes are illustrated. Right column: Topological fingerprint (bottom) for a protein (top). Image credit: Zixuan Cang.

been introduced to retain chemical and biological information during the topological abstraction [12]. This approach enciphers biological properties, such as hydrogen bonds, van der Waals interactions, hydrophilicity and hydrophobicity, into topological invariants, rendering a potentially revolutionary representation for biomolecules [13, 14].

Rational drug design is an imperative life science problem that ultimately tests out our understanding of biological systems. Designing efficient drugs for curing diseases is one of the most challenging tasks in biological sciences. Multicomponent persistent homology plays a crucial role in hot-spot prediction, drug pose analysis, binding affinity prediction, structure optimization, toxicity analysis and pharmacokinetic simulation. For example, the integration of machine learning with multiscale weighted colored graphs and multicomponent persistent homology provided the best free energy ranking for Set 1 (Stage 2) in D3R Grand Challenge 2, a world-wide competition in computer aided drug design (<http://users.math.msu.edu/users/wei/D3RFreeEnergy.pdf>).

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